

Operating Systems Lab

Week 2, 3, 4

Assumption : We have successfully installed a Linux distribution on our machine

Terminal, Terminal Emulator, Shell, Command Prompt, Commands

Directory System



Basic Linux Commands

Week 2

whoami (tells you about the username)

hostname (tells the name of host machine name) Usage: script filename

script (records terminal activity in a log file)

exit (if script is being recorded, HLT and that recording session) (also has a lot of other use cases, will discuss sometime)

cd (change directory) Usage: cd path

pwd (print/represent working directory)

ls (lists contents of a directory) ls path

ls -l (list contents of a directory but recursively) ls -l path

cat (concatenate, display contents of a file) cat file-path/file-name

touch (for creating empty files) [Usage: touch file-path/file-name.extension]

mkdir (for creating a directory)

mkdir -p (for creating nested directories)

rm (for removing files) [Usage: rm filepath]

rm -r (for removing directories) [Usage: rm filepath]

echo (takes some text as argument and simply output it) [Usage: echo Text]

Redirection Operators (redirects output of a command to a file)

> (Overwrite redirection operator; for redirecting output of a command into a file, it overwrites previous text in the file) [Usage: command > filename]

>> (Append redirection operator, almost same as > but it appends new text in the file) [Usage: command >> filename]

Pipe Operator

| (Pipe operator; redirects output of a command as input to another command) [Usage: command | command]

Wildcards (facilitate us in performing bulk actions on files and directories)

{} (Curly Braces Wildcard) (creates multiple copies of an argument/ filename)

{file..land} e.g {01..30} - {a..z}

{first-cop, second-cop, third-cop} e.g {2,5%,12%}, {rahg,abdu,abdullah,manzoor}

* (Asteric Wild Card)

pattern*pattern e.g f2023*section8 , all*malik, file*.txt

pattern* e.g Section-A*

pattern e.g *.txt , *.txt

tee (for reading some text as input writing it into a file, mostly used by piping it with other commands) [Usage: command | tee filename]

Week 3 & 4

cp (for copying files) Usage: [cp filepath new-path/new-filename]

cp -r (for copying directories) Usage: [cp -r directory-path new-path/new-name]

mv (for moving, renaming and hiding files/directories) [mv file-path new-path/new-name], [Usage: mv directory-path new-path/new-name]

grep [for searching patterns in a text has a very wide range of use-cases we used to simply search text in a file, it prints the line where particular results appear] Usage: [grep pattern filename] , [command | grep pattern]

grep -n (also print the line number when pattern found) Usage: [grep -n pattern filename]

grep -i (ignores upper/lower case of the pattern) Usage: [grep -i pattern filename] , [command | grep -i pattern]

grep -v (inverts the results of the pattern) Usage: [grep -v pattern filename], [command | grep -v pattern]

zip (for zipping files) zip path/filename original-file-path

zip -r (for zipping directories) zip -r path/filename original-directory-path

zip -m (for removing original copies) zip -m path/filename original-file-path

unzip (for extracting zip files) unzip original-zip-file-path/file-name.zip

unzip -d (for specifying directory) unzip -d directory-path/directory-name original-zip-file-path/file-name.zip

i

Keyboard Shortcuts

ctrl + alt + t (Opening the Terminal)

alt + F10 (Maximizing any window)

ctrl + l (for clearing the terminal screen, session activity can still be accessed by scrolling up)

TAB (For Autocomplete, very very useful and time saving feature)

Up Arrow (for copying/replicating previous command on your terminal)

Down Arrow (for next command, going to next command when you're copying previous commands with up arrow)

ctrl + a (for changing the position of the cursor and take it to the beginning of the command)

ctrl + e (for changing the position of the cursor and take it to the ending of the command)